Chapter 3 Checklist

Exponential and Logarithmic Functions

Ron Larson and Robert P. Hostetler, *Precalculus with Limits*, 2nd ed., p. 215. Tick each box as you go.

Work through the four parts in order: read each section, practise the matching Duolingo units, work the book-only topics that Duolingo does not cover, then finish the chapter. Duolingo units are grouped by lane (a grade or a topic in the app) and listed in the order the app shows them, so you can follow one lane from top to bottom. **full** = the unit practises the section's main skill, **partial** = part of it, **prereq** = a warm-up from an earlier grade.

In this chapter: Grade 11 covers exponentials and logarithms almost section by section. Duolingo stops at growth and decay, so the Gaussian, logistic, and logarithmic models of 3.5 come from the book only.

1. Read the textbook sections

- ☐ 3.1 Exponential Functions and Their Graphs — p. 216
- ☐ 3.2 Logarithmic Functions and Their Graphs — p. 227
- ☐ 3.3 Properties of Logarithms — p. 237
- ☐ 3.4 Exponential and Logarithmic Equations — p. 244
- ☐ 3.5 Exponential and Logarithmic Models — p. 255

2. Practise on Duolingo

Pick one lane and work down it; the grade lanes go deepest. The note after each unit gives its rating and the textbook sections it serves.

Grade 9

- ☐ Exponential function tables *prereq, 3.1* — unit 70
- ☐ Exponential function graphs *prereq, 3.1* — unit 71
- ☐ Exponential function equations *prereq, 3.1* — unit 72

Grade 11

- ☐ Exponential function tables *full, 3.1* — unit 38
- ☐ Exponential function graphs *full, 3.1* — unit 39
- ☐ Exponential function equations *full, 3.1* — unit 40
- ☐ Exponential growth *full, 3.1, 3.5* — unit 41
- ☐ Exponential decay *full, 3.1, 3.5* — unit 42
- ☐ Intro to the natural base *full, 3.1* — unit 48
- ☐ Continuous growth models *full, 3.1, 3.5* — unit 49
- ☐ Intro to logarithms *full, 3.2* — unit 50

- ☐ Logarithm value *full, 3.2* unit 51
- ☐ Product rule *full, 3.3* unit 52
- ☐ Quotient rule *full, 3.3* unit 53
- ☐ Power rule *full, 3.3* unit 54
- ☐ Change of base formula *full, 3.3* unit 55
- ☐ Intro to logarithmic functions *full, 3.2* unit 56
- ☐ Logarithmic functions in base 10 *full, 3.2* unit 57
- ☐ Logarithmic functions in base 2 *full, 3.2* unit 58
- ☐ Logarithmic function graphs *full, 3.2* unit 59
- ☐ Logarithmic function intercepts *full, 3.2* unit 60
- ☐ The natural logarithm *full, 3.2* unit 61
- ☐ Natural logarithm graphs *full, 3.2* unit 62
- ☐ Solve exponential equations using logarithms *full, 3.4* unit 63
- ☐ Solve base e equations using logarithms *full, 3.4* unit 64

Grade 12

- ☐ Properties of logarithms *full, 3.3* unit 15
- ☐ Undo exponentials *partial, 3.2, 3.4* unit 16

Algebra

- ☐ Logarithms *full, 3.2* unit 18
- ☐ Product and quotient rules *full, 3.3* unit 19
- ☐ Properties of logarithms *full, 3.3* unit 20
- ☐ Solve logarithmic and exponential equations *full, 3.4* unit 21

Algebraic Graphing

- ☐ Intro to exponential functions *full, 3.1* unit 15
- ☐ Exponential graphs and equations *full, 3.1* unit 16
- ☐ Exponential growth and decay *partial, 3.5* unit 18
- ☐ The natural base, e *full, 3.1* unit 19
- ☐ Logarithmic functions *full, 3.2* unit 20
- ☐ Logarithmic function graphs *full, 3.2* unit 21
- ☐ Natural logarithmic function *full, 3.2* unit 22

3. Section by section

For each section: what Duolingo covers (the units are ticked off in part 2) and the book-only topics to work in the textbook. Tick each book-only topic once you have worked examples of it.

3.1 Exponential Functions and Their Graphs p. 216

Grades 9 and 11 both cover exponential functions; grade 11 adds the natural base and continuous growth. Compound interest formulas are book only.

Duolingo: **Grade 9:** 70. Exponential function tables (prereq); 71. Exponential function graphs (prereq); 72. Exponential function equations (prereq)
Grade 11: 38. Exponential function tables (full); 39. Exponential function graphs (full); 40. Exponential function equations (full); 41. Exponential growth (full); 42. Exponential decay (full); 48. Intro to the natural base (full); 49. Continuous growth models (full)
Algebraic Graphing: 15. Intro to exponential functions (full); 16. Exponential graphs and equations (full); 19. The natural base, e (full)

Book only:

- ☐ Compound interest, n times per year and continuously

3.2 Logarithmic Functions and Their Graphs p. 227

Grade 11 covers logarithms, their graphs, and the natural logarithm nearly one to one with the section.

Duolingo: **Grade 11:** 50. Intro to logarithms (full); 51. Logarithm value (full); 56. Intro to logarithmic functions (full); 57. Logarithmic functions in base 10 (full); 58. Logarithmic functions in base 2 (full); 59. Logarithmic function graphs (full); 60. Logarithmic function intercepts (full); 61. The natural logarithm (full); 62. Natural logarithm graphs (full)
Grade 12: 16. Undo exponentials (partial)
Algebra: 18. Logarithms (full)
Algebraic Graphing: 20. Logarithmic functions (full); 21. Logarithmic function graphs (full); 22. Natural logarithmic function (full)

3.3 Properties of Logarithms p. 237

Fully covered: product, quotient, and power rules plus change of base in grade 11, consolidated in grade 12.

Duolingo: **Grade 11:** 52. Product rule (full); 53. Quotient rule (full); 54. Power rule (full); 55. Change of base formula (full)
Grade 12: 15. Properties of logarithms (full)
Algebra: 19. Product and quotient rules (full); 20. Properties of logarithms (full)

3.4 Exponential and Logarithmic Equations p. 244

Exponential equations are covered well; logarithmic equations and checking for extraneous solutions are lighter in the app.

Duolingo: **Grade 11:** 63. Solve exponential equations using logarithms (full); 64. Solve base e equations using logarithms (full)
Grade 12: 16. Undo exponentials (prereq)
Algebra: 21. Solve logarithmic and exponential equations (full)

Book only:

- ☐ Logarithmic equations
- ☐ Checking for extraneous solutions

3.5 Exponential and Logarithmic Models p. 255

Only exponential growth and decay models are covered. Gaussian, logistic, and logarithmic models are book only.

Duolingo: **Grade 11:** 41. Exponential growth (partial); 42. Exponential decay (partial); 49. Continuous growth models (partial)
Algebraic Graphing: 18. Exponential growth and decay (partial)

Book only:

- ☐ Gaussian models
- ☐ Logistic growth models
- ☐ Logarithmic models (pH, Richter scale, decibels)

4. Finish the chapter

- ☐ Chapter Summary p. 268
- ☐ Review Exercises p. 270
- ☐ Chapter Test p. 273
- ☐ Cumulative Test for Chapters 1-3 p. 274
- ☐ Proofs in Mathematics p. 276
- ☐ Problem Solving p. 277

Duolingo unit numbers are each unit's position in that grade's or topic's path in the Duolingo Math app, as recorded on 2026-10-07 (iOS); the app may reorder them. Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.